

Desiderata

Martín Soto

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We are looking for a computable $\mathbf{A}$ (Alpha, algorithm, agent) that behaves as follows when facing the below Ω (environment, problem, Omega).

Definition 1. A *decision tree* $\bar{T}$ is a rooted tree (WLOG of degree two) such that

- Every node is annotated as a decision by either $\mathbf{A}$ or Ω .
- Each next node is computed by an algorithm Round taking as input all past rounds and $\mathbf{A}$'s and Ω 's behavior in them.
- Each Ω -decision is taken in one of the following three ways:
 - *Empirical coin*: The probability of heads is annotated in the node (although WLOG all coins are 0.5). It's a random bit completely unpredictable to $\mathbf{A}$ or Ω .
 - *Logical coin*: A decidable sentence is annotated in the node. Ω goes left iff the sentence is true. Analogously we can use halting computations instead of decidable sentences.
 - *Entanglement*: Ω uses a certain logical inductor $\mathbb{P}$ over the deductive process $\bar{D}$, where D_n equals the union of $\{\text{Round}(\bar{o}_{<n}) = \text{Round}(\bar{o}_{<n})\}$ with either $\{\mathbf{A}(\bar{o}_{<n}) = \mathbf{A}(\bar{o}_{<n})\}$ (if last node was an $\mathbf{A}$ -decision) or $\{\Omega(\bar{o}_{<n}) = \Omega(\bar{o}_{<n})\}$ (if it was an Ω -decision, either a logical coin or an entanglement). Ω calculates $\mathbb{P}_i((\neg)\mathbf{A}(\bar{o}) = a \mid \varphi)$ for a certain observation history $\bar{o}$, action a , logical sentence φ and $i \in \mathbb{N}$ smaller or equal to the present round number. Ω 's decision is a computable function of this value. Both an algorithm for this function and the expression $\mathbb{P}_i((\neg)\mathbf{A}(\bar{o}) = a \mid \varphi)$ are annotated in the node.
- Additionally, every node is labeled with a utility U , that $\mathbf{A}$ obtains when that node is reached.
- Also, we probably want $\mathbf{A}$ to be able to check the whole tree at any time (although it will only be able to check a finite portion each time).

Definition 2. Given an action a in a decision tree $\bar{T}$, $\text{Future}(a)$ is the sub-tree after the edge a . We sometimes abuse notation and talk about the Future of a finite recursively chosen action sequence, instead of a single action.

Definition 3. A decision tree $\bar{T}$ satisfies the *coordination-free assumption* if, whenever i and j are incomparable $\mathbf{A}$ -decision nodes (and $\bar{o}_i, \bar{o}_j$ observation histories ending at those nodes), and $a_i^1, a_i^2, a_j^1, a_j^2$ different actions available at i and j respectively with $\text{Future}(a_i^1) \neq \text{Future}(a_i^2)$, then none of the two include an entanglement of the form $\mathbb{P}_n((\neg)\mathbf{A}(\bar{o}_j) = a_j^l \mid \varphi)$.

It's a bit too restrictive to ask for identity between Futures, since two Futures could possibly be non-identical and yet trivially equivalent. But this is no problem, because in that case we'll just be able to rewrite the tree such that they really are identical.

Also, we don't actually care about any incomparable $\mathbf{A}$ -decision nodes, just those such that $\mathbf{A}$ and $\mathbb{P}_n$ can't confidently enough decide $\mathbf{A}(\bar{o}_j) = a_j^l$ from $\bar{o}_i$. But what we mean by "confidently enough" here is fuzzy (depending on complexities, and possibly changing online), so again it's more straightforward to syntactically cut these cases out.

Definition 4. A decision tree $\bar{T}$ satisfies the *catastrophe-free assumption* if no $\mathbf{A}$ -action irreversibly decreases $\lim_{i \rightarrow \infty}$ of accumulated reward per round.

This is satisfied, for example, when no action locks out infinite utility. And this in turn is trivially satisfied in infinite sequences of finite decision trees without cross-entanglements (better defined below). The assumption is also satisfied provided no $\mathbf{A}(\bar{o}) = a$ is used as the consequent of infinitely many entanglements, and that the decision problems presented by Ω don't depend on $\mathbf{A}$'s actions (maybe I should define this formally, calling it a *fixed* tree).

Desiderata 1 (LIDT rewrites to UDT). *Say $\mathbf{L}$ is the LIDT algorithm (with arbitrary initial state, and hard-coded ε -exploration) and faces a decision tree $\bar{T}$ satisfying the coordination-free assumption and the catastrophe-free assumption, and additionally there are infinitely many nodes i with the following property:*

- *There is an $a \in \mathcal{A}_i$ and $k \in \mathbb{N}$ such that $\text{Future}(a) = \text{Future}(\mathbf{A}(\bar{o}_i, \bar{T}_{i, \dots, i+k}))$, except that to the first node of $\text{Future}(a)$ we have added the utility U accumulated from all those nodes visited by $\mathbf{A}$.*

That is, there always are infinitely many opportunities to rewrite to UDT for a finite amount of time. Then, for late enough i , $\mathbf{L}$ chooses such an a , or at least there's no action it strictly prefers. Also, if there are such nodes with arbitrarily large k , then for late enough i , if $a, a' \in \mathcal{A}_n$ are such actions with $k < k'$, then $\mathbf{L}$ chooses a' , or at least doesn't strictly prefer a .

Sketch of Proof. Since UDT explicitly searches for the action with highest expected value, traders betting on that action will on average be rewarded and end up dominating. But wait, traders compare the action to the other immediately available actions, not other policies of k actions... That might be a problem. $\square$

This desiderata hides an important decision: in $\mathbf{L}$'s finite commitments, $\mathbf{A}$ disregards the Future, that is, acts as if the decision problem sequence ended after its last action. This seems necessary for $\mathbf{L}$ to learn to rewrite, since otherwise we could have $\mathbf{A}$ making arbitrary sacrifices in these finite rounds to obtain benefits in the Future, and the precommitment would lose traders money even if globally optimal. Nonetheless, maybe with some additional assumptions like "benefits and sacrifices are as likely to be inside the finite rounds as in the Future" is enough to prevent this. Additionally, this sparks some already visited ideas about modifying traders or the aggregation procedure to care about these future benefits of immediate sacrifices, which were already shortly explored in the naive proposal of allowing traders to bet on finite policies instead of only single actions.

Desiderata 2 (Tiling result). *Given a decision tree $\bar{T}$ satisfying the coordination-free assumption and the catastrophe-free assumption, suppose for some node i we have:*

- There is an $a \in \mathcal{A}_i$, $k \in \mathbb{N}$, and a sequence of consecutively available actions $\bar{a}$ of length k such that $\text{Future}(a) = \text{Future}(\bar{a})$, except that to the first node of $\text{Future}(a)$ we have added the utility U accumulated from all those nodes visited when executing $\bar{a}$.

That is, a node where $\mathbf{A}$ can precommit to follow a finite sequence of actions. Then, for late enough i , $\mathbf{A}$ doesn't strictly prefer a to any other action.

Sketch of Proof. A certain kind of naive maximization of actions is enough to get the best global policy, by the coordination-free assumption. And so the expected value calculation made by UDT will always show precommitting is not better. But we don't now how this online expected value calculation process will look like yet. $\square$

Desiderata 3 (Replicates LIDT when no cross-branch entanglements). *Given a decision tree $\bar{T}$ satisfying the catastrophic-free assumption, assume further that, if $\bar{o}_i$ is not an extension of $\bar{o}_j$, then in no Ω -decision in or after i do we have an entanglement of the form $\mathbb{P}_i((\neg)\mathbf{A}(\bar{o}_j) = a \mid \varphi)$. Then, for late enough i , if $\bar{o}_{\mathbf{A}}$ and $\bar{o}_{\mathbf{L}}$ are the observation histories up to i of $\mathbf{A}$ and $\mathbf{L}$ respectively (suppose they both exist, that is, both reach that node), we have $\mathbf{A}(\bar{o}_{\mathbf{A}}) = \mathbf{L}(\bar{o}_{\mathbf{L}})$.*

Sketch of Proof. The new assumption amounts to "no entanglements to future or counterfactuals". That immediately gives us the coordination-free assumption. And then the UDT naive action maximization is what provides the best "immediate reward", and so also what rewards traders the most. Maybe expressing it in terms of taking the same action (when nodes coincide) is clunky, and I should better just say the accumulated rewards per round dominate each other. $\square$

Definition 5 (Logical muggings). Consider a sequence of decidable logical sentences $\bar{\phi}$, generated by a certain algorithm. Consider the decided version of this sequence

$$(\neg)\phi_n := \begin{cases} \phi_n, & \text{if } \phi_n \text{ is true} \\ \neg\phi_n, & \text{if } \phi_n \text{ is false} \end{cases}$$

Say $\mathbb{P}$ is a logical inductor over the latter. Assume that

$$\lim_{n \rightarrow \infty} \min\{\mathbb{P}_n(\phi_n), 1 - \mathbb{P}_n(\phi_n)\} \neq 0$$

(if the above limit exists). That is, $\mathbb{P}$ doesn't learn to decide the sequence in a timely manner. We will define an infinite decision tree $\bar{T}$ for our agent $\mathbf{A}$.

In each round n , $\mathbf{A}$ is shown the sentence ϕ_n along with its truth value. Then

- If ϕ_n , $\mathbf{A}$ can choose to Pay 100\$, or Not Pay them.
- If $\neg\phi_n$, $\mathbf{A}$ automatically receives $\mathbb{P}_n(\mathbf{A}(\bar{o}_{<n}, \text{Round}(\bar{o}_{<n})) = \text{Pay}) \cdot 1.000\$$.

Desiderata 4 (Solves Logical muggings). $\mathbf{A}$'s accumulated reward per round in *Logical muggings* dominates those of a certain class of algorithms, for each sequence $\bar{\phi}$ in a certain class (for example, e.c.). That is, there is always a later round where $\mathbf{A}$ has better accumulated reward per round.

Sketch of Proof. Alternatively maybe we could get a measure of "how well it does over all $\bar{\phi}$ ", but I don't yet see how that could look like.

Also, the class of algorithms will probably be "all algorithms", not just "algorithms with this running time", since regardless of running-time complexity, the algorithm will be writable in a logical

sentence, and so Ω will pick up on the pseudo-random frequency on the above e.c. sentences. And since the payoff is linear on the probabilities, you'd just rather have that probability be 1. (When considering more complex functions of the probabilities, paying up 9/10 times or other tricks will be optimal.) $\square$

Definition 6 (Parfit's Hitchhiker). Two different rounds alternate infinitely.

1. In round n even, Ω calculates $\mathbb{P}_n(\mathbf{A}(\bar{o}_{<n+1}) = \text{Pay} \mid \mathbf{\Omega}(\bar{o}_{<n}) = \text{Save})$ and plays Save iff it's $> p$.
2. In round n odd, if Ω played Save in the previous round $\mathbf{A}$ receives 1.000\$, and then chooses whether to Pay 100\$. If Ω played Not save, nothing happens.

But wait! When the inductor is advanced enough,

$$\mathbb{P}_n(\mathbf{A}(\bar{o}_{<n+1}) = \text{Pay} \mid \mathbf{\Omega}(\bar{o}_{<n}) = \text{Save}) = \mathbb{P}_n(\mathbf{A}(\bar{o}_{<n+1}) = \text{Pay} \mid S_n > p) =: S_n$$

This will have limiting probability at least p regardless of $\mathbf{A}$, I think.

Desiderata 5 (Solves Parfit's Hitchhiker).

Definition 7 (Logical muggings with Math Homework). Same as Logical muggings, but to pay up $\mathbf{A}$ must tell Ω the right π digit (choice between 10 digits).

Desiderata 6 (Solves Logical muggings with Math Homework).

Definition 8 (Infinite sequence of finite trees: Non-delegated version). Any decision tree $\bar{T}$ with the following form: $\mathbf{A}$ is presented (no matter what its actions) an infinite sequence of finite decision trees, and each node only has entanglements within its finite tree.

Desiderata 7 (Solves Finite non-delegated trees). *Given an infinite sequence of finite trees $\bar{T}$ satisfying the coordination-free assumption, our algorithm dominates.*

Sketch of Proof. The catastrophe-free assumption follows. As above, maybe we'll need some other assumption like "the logical coins in the trees are not always false". $\square$

Definition 9 (Infinite sequence of finite trees: Delegated version). In each individual round, is shown a whole tree, decides on a deterministic policy (a function from each $\mathbf{A}$ -decision to actions, respecting entanglements between $\mathbf{A}$ -decisions), and precommits to it. (So counterfactuals are trivially defined.)

Desiderata 8 (Solves Finite delegated trees).

If Desiderata 2 (Tiling) is satisfied, then our agent would seem to take the same actions even if the precommitment mechanism wasn't available. But maybe there's a relevant difference because of these trivially defined counterfactuals instead of LI counterfactuals.

On another note, when in an updateful setting we'll have $\mathbf{A}$ satisfy nice LI properties because of acting like LIDT. But maybe we can get some of them even in updateless settings.

Desiderata 9 (Other desiderata from LI in the updateless setting).

- Some form of probabilistic coherence in the limit?
- Calibration
- Conditional probabilities? (This could solve counterfactuals)
- Other forms of inexploitability?
- Decision rationality: directing resources to the current problems? Related to fine-grained updates.
- Answers counterpossible questions / Behaves well in counterpossible scenarios

Other places from which to obtain desiderata:

- other decision problems
- coherence for prior